

6a	electric and magnetic fields <u>normal</u> to each other <u>in the same region</u> <i>either</i> charged particles <u>enters region normal</u> to both fields or <u>correct B direction wrt E</u> for zero deflection (in drawing) For <u>no deflection</u>, $v = E/B$ or <u>no net force</u>.	M1 A1 A1
6bi	magnetic force on ion in path B provides for centripetal force By N2L, $Bqv = m \frac{v^2}{r}$ $m = \frac{rBq}{v} = \frac{\frac{12.3}{2} \times 10^{-2} \times 640 \times 10^{-3} \times 1.6 \times 10^{-19}}{9.6 \times 10^4}$ $= 6.56 \times 10^{-26} \text{ kg}$ $= \frac{6.56 \times 10^{-26}}{1.66 \times 10^{-27}} = 40 \text{ u (or 39.5 u)}$	B1 C1 A1
6bii	Since the ions are of the <u>same isotope</u>, they all have the <u>same mass</u> regardless of the paths undertaken. Using the equation in answer to (b)(i), the <u>radius of the path is inversely proportional to q</u> (or state equation for r) Hence, the ions in path A have <u>thrice</u> the charge compared to ions in path B.	B1 B1 B1

7a	energy from 1 nucleus = $(1.77 \times 10^{13}) / (6.02 \times 10^{23})$	C1
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